

2016 Q31

$$\frac{mV^2}{r} = \frac{GMm}{r^2} \rightarrow V^2 = \frac{GM}{r}$$

$$r \rightarrow ar$$

$$\tilde{V}^2 = \underline{GM}$$

2021:

$$\begin{array}{c} 350 \text{ Hz} \\ \rightarrow \\ 80 \text{ km/h} \end{array}$$

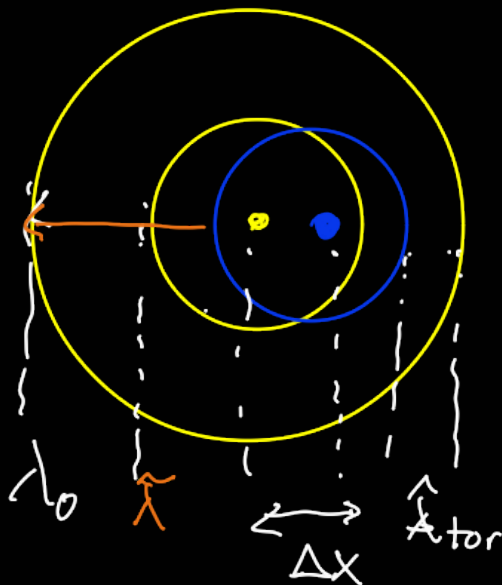
Q

$$\leftarrow 350 \text{ Hz}$$

$$V = f\lambda$$

$$V_s = f_s \lambda_s$$

$$V_s = \frac{\lambda_s}{T_s}$$



$$\hat{\lambda}_{\text{towards}} = \lambda_0 - \Delta x$$

$$\frac{\Delta x}{\Delta t} = V_t$$

$$\hat{\lambda}_a = \Delta x + \lambda_0 \text{ (away)}$$

$$\hat{\lambda}_{\text{towards}} = \lambda_0 - \Delta x = \lambda_0 - V_t \Delta T_t$$

$$\tilde{V}_{\text{towards}} = V_s T_s - \underline{V_t \Delta T_t}$$

Speed of sound is the same in all reference frames

$$V_s T_s = V_s T_s - V_t \Delta t$$

- Look at the case that we've moved $\Delta t = T_s$

$$V_s T_{\text{new}} = V_s T_s - V_2 T_s$$

$$V_s T_{\text{new}} = (V_s - V_2) T_s$$

$$f_{\text{new}} = \left(\frac{V_s}{V_s - V_2} \right) f_s \quad (\text{towards})$$

$$\lambda_{\text{new}} = (\lambda_1 - \Delta x_1) + (\lambda_2 - \Delta x_2)$$

$$= V_s(T_s)_1 - V_1 \Delta T_1 + V_s(T_s)_2 - V_2 \Delta T_2$$

$$V_s t_{\text{new}} = (V_s - V_1 + V_s + V_2) T_{\text{sound}}$$

$$V_s t_{\text{new}} = (2V_s + (V_2 - V_1)) t_{\text{sound}}$$

$$f_{\text{new}} = \frac{V_s}{(2V_s + (V_2 - V_1))} f_0$$

- $V_s = C_s = 340 \text{ m/s}$

- $f_{\text{new}} = 5 \text{ Hz}$

- $f_0 = 350 \text{ Hz}$

- $V_1 = 80 \text{ km/h}$

$$\frac{350}{5} = 80$$

$$2V_s - (V_2 + V_1) = 80 \text{ m/s}$$

$$V_2 = -78 \text{ m/s} + V_1$$

$$= -78(60) \text{ m/s} + V_1$$

Wind instruments:



$\underline{l = \frac{1}{2} \lambda_1}$

$\left. \begin{aligned} f \lambda &= V \\ f &= \frac{v}{2l} \end{aligned} \right\}$

B

• fundamental node



all others are harmonics

1st

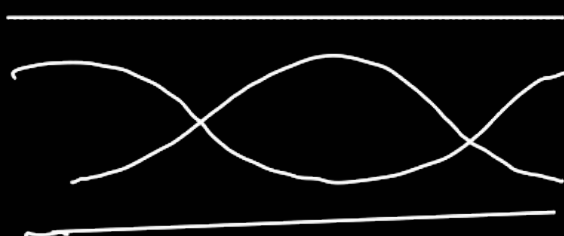


2nd



$\lambda_2 = \frac{1}{2} \lambda_1$

Second Harmonic:



$l = \lambda_1$

$\frac{V}{l} = f_2 = 2f_1$

$f_2 = 2f_1$

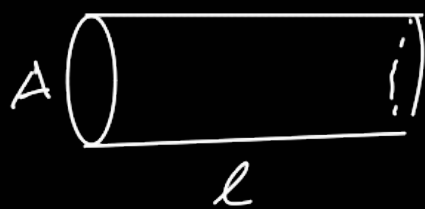
3rd Harmonic:

$f_3 = 3f_1$

$\underline{l = \frac{3}{2} \lambda_3}$

• $V = f_3 \lambda_3 = \frac{2}{3} f_3 l$

$f_3 = \frac{3}{2} \frac{V}{l} = 3 \left(\frac{V}{2l} \right) = 3f_1 \quad \checkmark$



Δp

$[X] = \frac{mVol}{\Delta t}$

$[M] = N s m^{-2}$

$P \sim \frac{F}{A} \sim N$

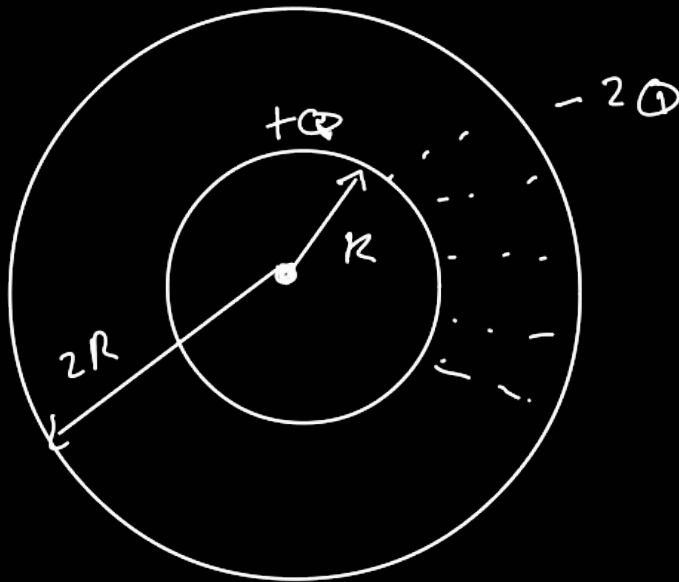
$$X(r, \frac{p}{e}, n) = \frac{\text{Vol}}{\Delta t} = \frac{p}{n \ell}$$

$$\frac{p}{n \ell} \rightarrow \frac{N n^{-2}}{N \text{ s m}^{-2}} \rightarrow \left[\frac{1}{\text{s m}} \right]$$

$$\rightarrow X = \left(\frac{\Delta p}{L} \right) \frac{D^4}{n}$$

$$X \propto \left(\frac{p}{e} \right) \frac{r^4}{n}$$

2023:



$$|x| < R: E = 0$$

Gauss' Law: $\Phi_E = EA \cos(\theta)$

$$\Phi_E = \sum_{\text{closed}} \vec{E} \cdot \vec{A} = \frac{Q_{\text{enc.}}}{\epsilon_0}$$

$$\therefore |x| < R \Rightarrow 0 = E$$

(i) $|x| < R$

$$\Phi_E = \sum_{\text{closed}} E_{\text{enc}} A = \frac{Q_{\text{enc}}}{\epsilon_0}$$
$$= E 4\pi R^2 = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 R^2}$$

(ii) $|x| > 2R$

$$\sum E_{\text{enc}} 4\pi R^2 = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E 4\pi R^2 = \frac{-Q}{\epsilon_0}$$

$$E = \frac{-Q}{4\pi R^2 \epsilon_0}$$

* Potential: $\vec{F} = qE = -\nabla V$

$$-\frac{q Q_{\text{tot}}}{4\pi\epsilon_0 R^2} \hat{r} = \nabla \phi$$

$$V_{(r)} = - \int_{\infty}^r \frac{q Q_i}{4\pi\epsilon_0 R^2} dR$$

$$= \frac{q Q_i}{4\pi\epsilon_0} \frac{1}{R} \Big|_{\infty}^r$$

$$= \frac{q Q_i}{4\pi R \epsilon_0}$$

iii) $Q_i = -Q$

$$\infty \longrightarrow 2R$$

$$V_{(r)} = \frac{Q^2}{4\pi R \epsilon_0}$$

$$4\pi\epsilon_0 R$$

$$\begin{aligned}
 (ii) \quad V_{(ii)} &= + \frac{Q Q_i}{4\pi\epsilon_0} \int_{2R}^R \frac{dR}{R^2} \\
 &= - \frac{Q Q_i}{4\pi\epsilon_0} \left(\frac{1}{R} - \frac{1}{2R} \right) \\
 &= - \frac{Q^2}{(4\pi\epsilon_0)(2R)}
 \end{aligned}$$

$$(i) \quad R \rightarrow 0 : \quad V = 0$$

$$V_{tot} = \frac{Q^2}{2(4\pi\epsilon_0 R)}$$

$$i\partial_t \psi = \left(\frac{p^2}{2m} + V(x) \right) \psi$$

$$\underline{p_\mu = -i\hbar \partial_\mu} : \quad p^\mu p_\mu = -\hbar^2 \nabla^2$$

accast.

$$\begin{aligned}
 \psi(x+a) &= \psi(x) + a^\mu \partial_\mu \psi + \mathcal{O}(a^2) \\
 &= e^{a^\mu \partial_\mu} ()
 \end{aligned}$$

$$= e^{i \frac{a^\mu}{\hbar} p_\mu} = e^{i \frac{\vec{a}^\mu p_\mu}{\hbar}}$$

$$p^\mu = m \frac{dx^\mu}{dt} = m \frac{dx^\mu}{dt} = p_\mu \quad \checkmark$$

$$i\partial_t \psi = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi$$

General V:

$$(I) \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + 0 \right) \psi = E \psi$$

$$\frac{\partial^2}{\partial x^2} \psi = - \underbrace{\frac{2mE}{\hbar^2}}_{k^2} \psi$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\psi_I(x) = A e^{ikx} + B e^{-ikx}$$

$$(II) \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V_0 \right) \psi = E \psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} = (E - V_0) \psi$$

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}} \quad \left| \quad k_2 = \sqrt{\frac{(E - V_0) 2m}{\hbar^2}} \right.$$

$$(i) \quad \psi_I(0) = \psi_{II}(0)$$

$$1 + \sqrt{R} = \sqrt{T \left(\frac{k}{k_1} \right)}$$

$$\frac{\partial \psi_I(0)}{\partial x} = \frac{\partial \psi_{II}(0)}{\partial x}$$

.....

$$(c) \left(-\frac{\hbar^2}{2m} \nabla_x^2 - \frac{e^2}{4\pi\epsilon_0|x|} \right) \psi(x) = E \psi(x)$$

$$\gamma_i = \sqrt{\frac{4\pi\epsilon_0}{e^2}} x_i$$

$$\left(-\frac{4\pi\epsilon_0\hbar^2}{2} \nabla_x^2 - \frac{meE}{|x|} \right) \psi = (4\pi\epsilon_0 meE) \psi$$

$$x_i = \alpha y_i \quad \sum y_i^2 = |y|^2 = \alpha^2 |x|^2$$

$$|y| = \alpha |x|$$

$$\frac{\partial}{\partial x_i} = \frac{\partial y_j}{\partial x_i} \frac{\partial}{\partial y_j} = \delta_i^j \alpha \frac{\partial}{\partial y_j}$$

$$\sum \left(\frac{\partial}{\partial x_i} \right)^2 = \sum \delta_i^i \alpha^2 \frac{\partial^2}{\partial y_i^2}$$

$$\left(-\frac{\alpha^{-2}}{2} \nabla_x^2 - \frac{1}{\alpha |x|} \right) \tilde{\psi} = \epsilon \tilde{\psi}$$

$$\left(-\frac{1}{2} \nabla_x^2 \quad \nabla_x^2 - \frac{\alpha}{|x|} \right) = \alpha^2 \epsilon \tilde{\psi}$$

$$\left(-\frac{\hbar^2}{2m} \nabla_x^2 + \frac{e^2}{4\pi\epsilon_0 |x|} \right) \psi(x) = E \psi(x)$$

$$|x|^2 = x_1^2 + x_2^2 + x_3^2$$

$$x_i = \alpha y_i \quad |x| = \alpha |y|$$

$$\bullet \quad \frac{\partial}{\partial x_i} = \frac{\partial y_j}{\partial x_i} \frac{\partial}{\partial y_j} = \alpha^{-1} \delta_i^j \frac{\partial}{\partial y_j}$$

$$= \alpha^{-1} \frac{\partial}{\partial y_i}$$

$$\left(\frac{\alpha^{-2}}{2} \frac{\hbar^2}{2m} \nabla_y^2 + \frac{e^2}{4\pi\epsilon_0 \alpha |y|} \right) \psi = E \psi$$

$$X(|x|) =$$

$$\left(\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - \frac{e^2}{4\pi\epsilon_0 |x|} \right) \psi = E \psi$$

$$\left(|x| - \frac{1}{2} \frac{\partial^2}{\partial x^2} \right) \psi = \frac{m}{\hbar^2} \left(E |x| + \frac{e^2}{4\pi\epsilon_0} \right) \psi$$

$$|x| = \frac{m}{\hbar^2} \left(E |x| + \frac{e^2}{4\pi\epsilon_0} \right)$$

$$\frac{1}{E} \left(\frac{\hbar^2}{m} |x| - \frac{e^2}{4\pi\epsilon_0} \right) = |x|$$

$$- \frac{1}{2} \left(\frac{\partial}{\partial x} \right)^2 - \frac{e^2}{4\pi\epsilon_0 |x|}$$

$$\text{lii) } \psi(y) = (e^{-|y|/c_0})$$

$$\textcircled{I} E = 10^{19} \text{ J}$$

$$m \frac{x}{t^2}$$

$$W = \int F \cdot \Delta x = 10^{19} = \frac{1}{2} m v^2$$

$$\text{kg} \frac{\text{m}^2}{\text{s}^2}$$

$$\text{J} : \text{kg} \frac{\text{m}^2}{\text{s}^2}$$

$$10^{19} = \frac{1}{2} m v^2$$

$$\sqrt{\frac{2(10^{19})}{m}} = v$$

$$\sqrt{\frac{2 \cdot 10^{19}}{250}} = \left(\right) \frac{\text{m}}{\text{s}}$$

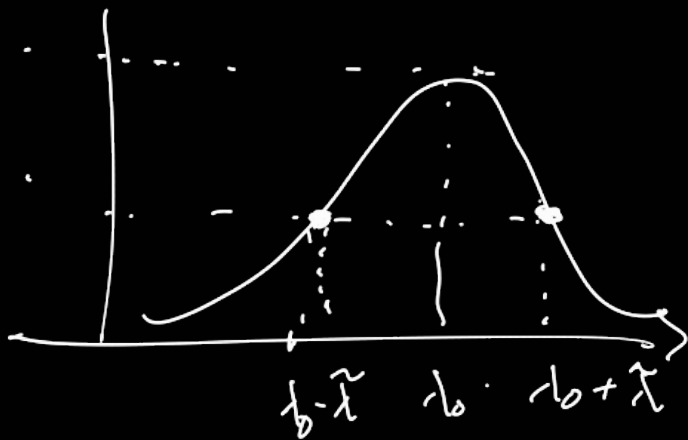
$$\lambda = 532 \text{ nm}, \quad \Delta t = 5 \times 10^{-16} \text{ s}$$

$$f(\lambda) = a e^{-\frac{(\lambda - \lambda_0)^2}{B}}$$

$$\left. \frac{\partial f}{\partial \lambda} \right|_0 = 0 \quad \lambda = \lambda_0 \text{ (maximum)}$$

$$f(\lambda) = a e^{-\frac{(\lambda - \lambda_0)^2}{B}}$$

$$B(t) = B_0 e^{-138 \left(\frac{t}{\tau}\right)^2}$$



$$f(\lambda_0 + \hat{\lambda}) = a e^{-\frac{(\lambda_0 + \hat{\lambda}) - \lambda_0)^2}{B}} = \frac{a}{2}$$

$$a e^{-\frac{\hat{\lambda}^2}{B}} = \frac{a}{2}$$

$$\frac{\hat{\lambda}}{B} = \ln(2)$$

$$\hat{\lambda} = B \ln(2)$$

$$(2) \quad (M_{\text{sys}})_\lambda = 2 \frac{\text{MeV}}{c^2} \quad | \quad U_1 E_1 = 3 \text{ MeV}$$

$$M_2 = 4 \text{ MeV}/c^2$$



$$\frac{V_0^2}{c^2} = 1 \quad \left(\frac{V_0}{c}\right) = 1$$

$$m_1 = 2 \frac{\text{MeV}}{c^2} \quad | \quad K.E = 3 \text{ MeV} = \frac{1}{2} m v^2$$

$$\left(\frac{V_0}{c}\right)^2 = 3 \Rightarrow \underline{V_0 = \sqrt{3} c}$$

$$m_1 V_0 = (m_1 + m_2) V_f$$

$$\frac{1}{2} m_1 \left(\frac{V_0}{c}\right)^2 = \frac{1}{2} (m_1 + m_2) V_f^2$$

$$2(\sqrt{3})^2 = (2 + 4) \left(\frac{V_f}{c}\right)^2$$

$$\frac{2 \cdot 3}{6} = \left(\frac{V_f}{c}\right)^2 = > 1?$$

$$2 \left(\frac{V_0}{c^2}\right) = 6 \left(\frac{V_f}{c^2}\right)$$

$$2 \sqrt{3} c = 6 V_f$$

$$V_f = \frac{\sqrt{3}}{3}$$

$$\delta N \delta t \sim \frac{1}{2}$$

$$\delta N = \left| -\frac{c}{\lambda^2} \right| \delta \lambda$$

$$\frac{c}{\lambda^2} \delta \lambda \delta t \sim \frac{1}{2} \Rightarrow \delta \lambda = \frac{\lambda^2}{2c \delta t}$$

$$\delta \lambda \approx \frac{(532 \times 10^{-9} \text{ m})^2}{2 (2 \times 10^8 \text{ m/s}) (8 \times 10^{-16} \text{ s})}$$

$$\frac{(532)^2 \cdot 10^{-18}}{6 \times 10^{-8}} = \frac{(532)^2}{30} \times 10^{-10} \text{ m/s}$$

$$= \frac{500}{(5 \cdot 10^2)^2} \cdot \frac{5 \cdot 10^3}{6} \text{ nm}$$

$$\delta \lambda \quad \cancel{.80}$$

$$.6(10^3)$$

$$\approx 600 \text{ nm}$$

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\Delta E = \left| \frac{\hbar c}{\lambda^2} \right| \Delta \lambda$$

$$\left| \frac{\hbar c}{\lambda^2} \right| \Delta \lambda \Delta t \geq \frac{\hbar}{2}$$

$$\Delta \lambda \geq \frac{\lambda^2 \hbar}{\Delta t \cdot 2 \hbar c} = \frac{\lambda^2}{2 \pi \Delta t \cdot 2 \hbar c}$$

$$\Delta \lambda = \frac{\lambda^2}{4 \pi \Delta t c} = \frac{(532 \times 10^{-9} \text{ m})^2}{4 \pi (5 \times 10^{-16} \text{ s})(3 \times 10^8 \text{ m/s})}$$

$$= \frac{(532)^2 (10^{-18}) \text{ m}^2}{60 \pi \cdot 10^{-8} \text{ m}}$$

$$= \frac{(532)^2}{60 \pi} 10^{-10} \text{ m}$$

$$= \frac{(500 + 32)^2}{6 \pi} \text{ nm}$$

$$\underline{283024}$$

$$= 11320 \text{ nm}$$

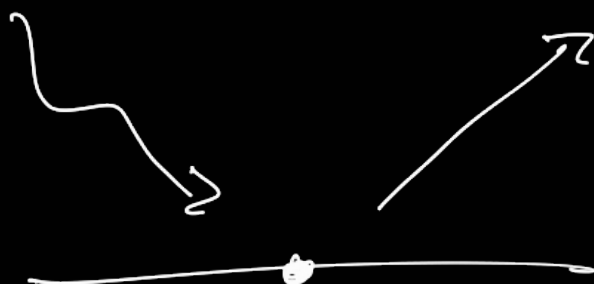
(i)

$$E_i = 3 \text{ MeV} + 4 \text{ MeV} = \underline{7 \text{ MeV}}$$

$$E_f = (m_1 + m_2) c^2 + \frac{1}{2} (m_1 + m_2) v^2$$

$$\sqrt{\frac{2}{6}} = \frac{1}{\sqrt{3}}$$

$$E = 0.5 \text{ eV} \quad \lambda = 500 \text{ nm}$$



$$W = h\nu - K$$

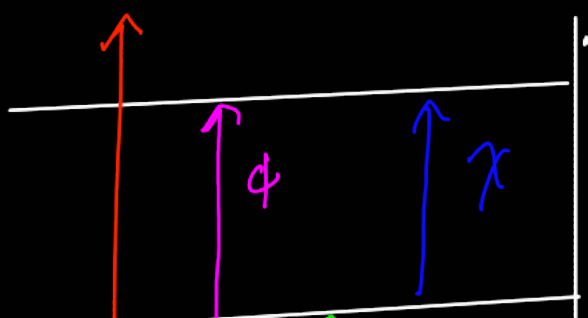
$$\begin{cases} E_i = h\nu - W \\ E_f = K \end{cases}$$

$$W_1 = h\nu_1 - \underline{K_1}$$

$$W_2 = h\nu_2 - \underline{K_2}$$

$$|\Delta| = |W_1 - W_2| = h|\nu_1 - \nu_2| = h|1001 \text{ nm}|$$

Semi-conductor:

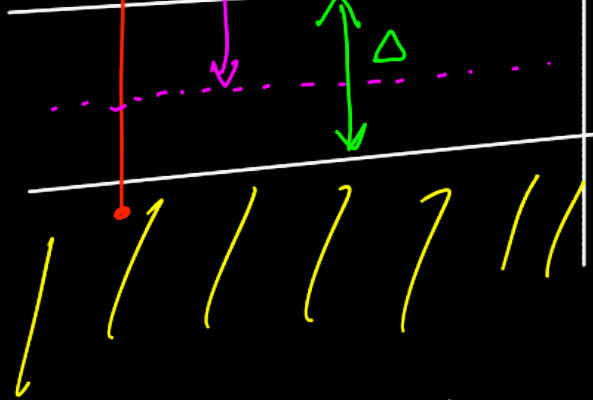


ϕ : Work fn

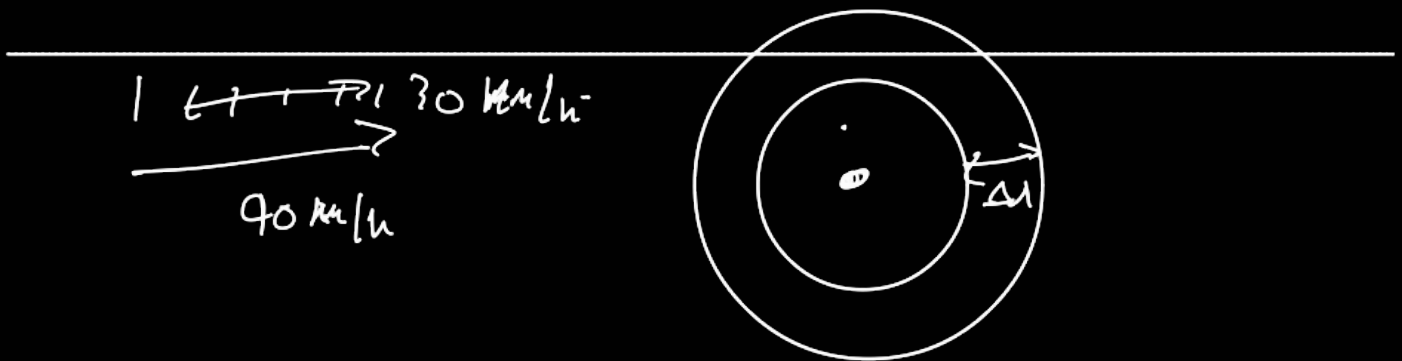
Δ : band gap

χ : electron aff.

$(\Delta + \chi)$: photoemission



$$h\nu - \phi = (E_e)$$



Ref frame: Change so that obs. is moving with the wind

$$\Rightarrow \text{Given: } V_{\text{siren}} = -30 \text{ km/h}$$



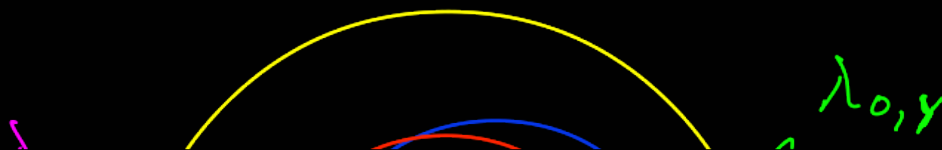
$$V = \lambda f$$

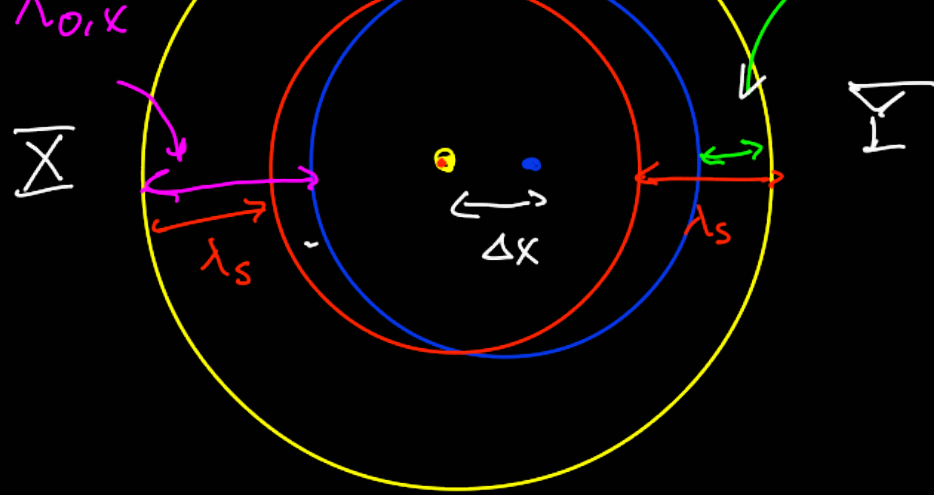
$$V = \frac{\lambda}{T}$$

$$\tilde{\lambda} = \lambda_0 - \Delta x$$

$$\tilde{\lambda} = V -$$

Moving Source:





$$\begin{cases} \Delta x = \lambda_s - \lambda_{oy} \\ \Delta x = \lambda_{ox} - \lambda_s \end{cases}$$

$$\Delta x = v \Delta t$$

$$\begin{cases} \lambda_{oy} = \lambda_s - \Delta x = \lambda_s - v \Delta t \\ \lambda_{ox} = \lambda_s + \Delta x = \lambda_s + v \Delta t \end{cases}$$

Δt : period of emission

$$v_s = \frac{\lambda_s}{\Delta t_s}$$

$$\begin{cases} \lambda_{oy} = (v_s - v) \Delta t_o \\ \lambda_{ox} = (v_s + v) \Delta t_o \end{cases}$$

$$\lambda_{oy} = v_s \Delta t_y$$

$$\lambda_{ox} = v_s \Delta t_x$$

$$v_s \Delta t_y = (v_s - v) \Delta t_o$$

$$\lambda_s \Delta t_k = (v_s + v) \Delta t_o$$

$$f_x = \left(\frac{v_s}{v_s + v} \right) f_s$$

$$f_y = \left(\frac{c_s}{c_s - v} \right) f_s$$

$$\Delta x = \lambda_s - \lambda_o \pm \Delta y$$

$$(v \Delta t_s \mp v_y \Delta t_f) = v_s \Delta t_s - \lambda_o$$

$$\bullet \lambda_o \pm v_y \Delta t_f = v_s - v$$

$$\lambda_o \pm v_y \Delta t_f = v_s - v$$

$$v_s \Delta t_f \pm v_y \Delta t_f = (v_s - v) \Delta t_s$$

$$(v_s \pm v_y) \Delta t_f = (v_s - v) \Delta t_s$$

$$f_f = \frac{(v_s \mp v_y)}{v_s - v} f_o \leftarrow \text{opposite}$$

$+$: moving away
 $-$: moving towards

$$f_f = \left(\frac{v_s - v_y}{v_s - v} \right) f_o$$

Source towards / moving away

* Move ref. frames:

* wind at rest

car: 60 km/h , Shen: $V_{\text{shen}} = -30 \text{ km/h}$

$$f_o = f_s \left(\frac{c_s + v_o}{c_s - v_s} \right)$$

$$f_o = f_s \left(\frac{c_s + 0}{c_s - 30} \right)$$

$$F = mg = m \frac{d^2 y}{dt^2}$$

$$F = \frac{-M G}{R^2} = m \frac{d^2 r}{dt^2}$$

$$\frac{M(R) G}{R^2}$$

$$M(R) = \rho \frac{4}{3} \pi R^3$$

$$- \left(\rho_{\text{const}} \right) \frac{4}{3} \pi r G = \frac{d^2 r}{dt^2}$$

$$- r \left(\frac{M_{\oplus} G}{R_{\oplus}^3} \right) = - \frac{r g}{R_{\oplus}} = \frac{d^2 r}{dt^2}$$

$$r(t) \sim A e^{i \omega t}$$

$$\omega^2 = \frac{g}{R_{\oplus}}$$

$$\omega = (2\pi f)^2 = \sqrt{\frac{g}{R_{\oplus}}}$$

$$2\pi \sqrt{\frac{R_{\oplus}}{g}} = T$$



$$\frac{\pi R}{T} = \frac{D}{T} = \text{Vel.}$$

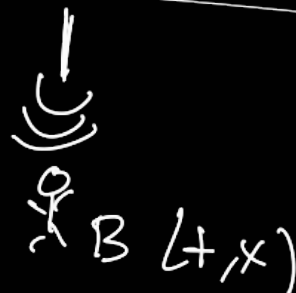
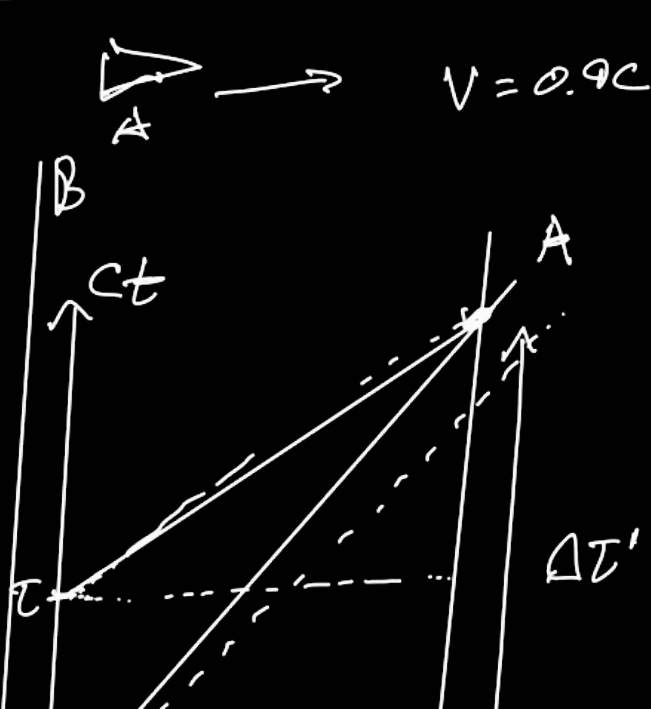
$$\frac{\pi R}{2\pi \sqrt{\frac{R\phi}{g}}} = \text{Vel}$$

$$\frac{\sqrt{R\phi g}}{2} = \text{Vel}$$

$$\frac{1}{2} m v^2 = \left(\frac{G M_{\oplus} R}{R R} \right) g$$

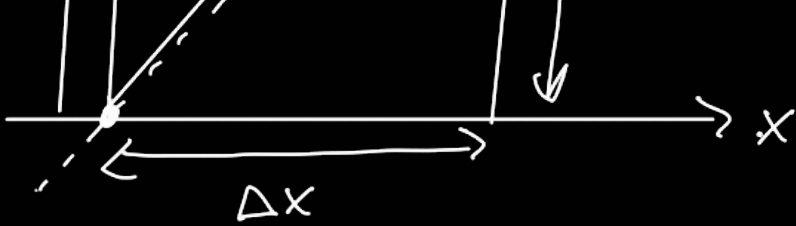
$$V = \sqrt{2 g R} = \sqrt{2} \sqrt{g R}$$

no



$$V = \frac{\Delta x}{\Delta t}$$

$$\frac{\Delta t}{\Delta x} = \frac{1}{0.9c}$$



$$\frac{c(\Delta\tau' - \tau)}{\Delta x} = \frac{1}{c} \quad \left| \quad \frac{c\Delta\tau'}{\Delta x} = \frac{1}{0.9c} \right.$$

$$\Delta x = 0.9c^2 \Delta\tau' = c^2(\Delta\tau' - \tau)$$

$$\tau = \frac{1}{10} \Delta\tau' \Rightarrow \Delta\tau' = 10\tau$$

$$f' = \left(\frac{1+B}{1-B} \right)^{1/2} f$$

$$B = \frac{v}{c} = 0.9$$

$$\sqrt{19} f$$

$$ds^2 = -dt^2 + dx^2$$

$$-c^2 d\tau^2 = -c^2 dt^2 + dx^2$$

$$-1 = -\frac{dt^2}{d\tau^2} + v^2$$

$$d\tau^2 = dt^2 - \frac{dx^2}{c^2}$$

$$\frac{d\tau^2}{dt^2} = 1 - \frac{v^2}{c^2} =$$

$$\left(\frac{d\tau}{dt} \right)^2 = 1 - \frac{v^2}{c^2}$$

$$d\tau = \sqrt{1 - \frac{v^2}{c^2}} dt$$

$$\frac{dx}{dt} = 1 \quad 1 - \frac{v^2}{c^2} = 1$$

$$-c dt^2 + dx^2 = -c dt'^2 + dx'^2$$

$$-\frac{dt}{dt'} + \left(\frac{dx}{dx'}\right)^2 = -1 + \left(\frac{dx'}{dx}\right)^2$$

$$-c^2 \gamma^2 dt^2 + dx^2 = -c dt'^2 + dx'^2$$

$$c^2 \gamma^2 (-c^2 dt^2 + dx'^2) + dx^2 =$$

$$dx^2 = (1 - c^2 \gamma^2) dx'^2$$

$$\left(1 - \frac{c^2}{1 - \frac{v^2}{c^2}}\right) = \left(\frac{1 - \frac{v^2}{c^2} - c^2}{1 - \frac{v^2}{c^2}}\right)$$

$$\Delta \tau^2 = \left(1 - \frac{v^2}{c^2}\right) \Delta t^2$$

$$\gamma \Delta \tau = \Delta t$$

$$\Delta \tau^2 = \left(1 - \frac{9^2}{10^2}\right) \Delta t^2$$

$$\frac{100 - 81}{100}$$

$$\Delta \tau = \sqrt{\frac{19}{100}} \Delta t$$

$$x^\mu = (t, x^i)$$

$$\frac{dx^\mu}{dt} = \left(\frac{dt}{dt}, \frac{dx^i}{dt}\right)$$

$$d\tau = (1, v^i)$$

$$\frac{dx^\mu}{d\tau} \frac{d\tau}{dt} = \gamma^{-1}$$

$$\frac{dx^\mu}{d\tau} = \gamma (1, v^i)$$

$$V^\mu = \gamma (1, v^i)$$

$$x'^\mu = (t', x'^i)$$

$$x^\mu = \frac{\partial x^\mu}{\partial x'^\sigma} x'^\sigma$$

$$= \frac{\partial}{\partial x'^\sigma}$$

$$x^0 = c \frac{\partial t}{\partial x'^\sigma} x'^\sigma$$

$$= - \frac{\partial t}{\partial t'} c t' + \frac{\partial t}{\partial x^i} x^i$$

$$= \frac{\partial t}{\partial t'} \left(-c t' + \left(\frac{\partial x^i}{\partial t'} \right)^{-1} x^i \right)$$

$$c t = \frac{\partial t}{\partial t'} \left(-c t' + \frac{x^i}{v^i} \right)$$

$$= \frac{\partial t}{\partial t'} \left(-t' + \frac{x'}{cV'} \right)$$

$$X = - \frac{\partial X}{\partial t'} t' + \frac{\partial X}{\partial x'} x'$$

$$= \frac{\partial X}{\partial t'} \left(-t'' + V^{-1} x'' \right)$$

$$= \frac{\partial X}{\partial \tau} \frac{\partial \tau}{\partial t'}$$

$$= \gamma V \gamma^{-1}$$

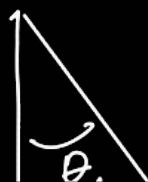
$$= (x' - V t'')$$

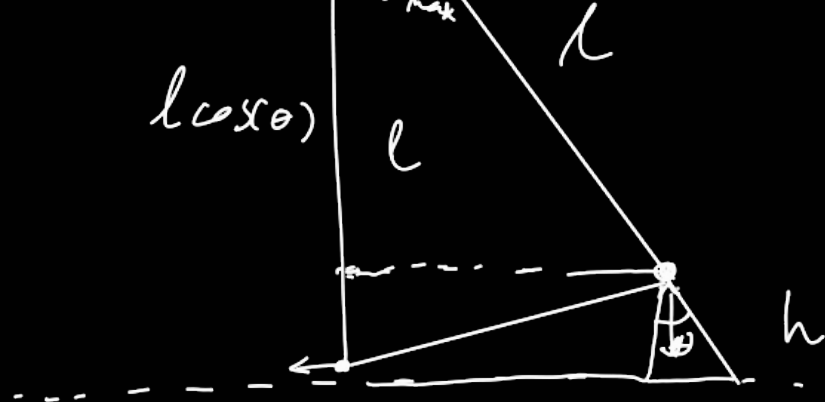
$$X'' = \frac{\partial X''}{\partial x''} x''$$

$$x' = - \frac{\partial x}{\partial (ct')} (ct') + \frac{\partial x}{\partial x'} dx'$$

$$\frac{\partial x}{\partial \tau} \left(- \frac{\partial \tau}{\partial t} t' + \frac{\partial \tau}{\partial x'} dx' \right)$$

$$- \gamma^{-1} t' +$$





$$U = mgh = \frac{1}{2} m V_t^2$$

$$V_t = \omega l$$

$$\omega = \frac{d\theta}{dt}$$

$$mgh = \frac{1}{2} m \left(\frac{d\theta}{dt} l \right)^2$$

$$l \cos(\theta) + h = l$$

$$h = l(1 - \cos(\theta))$$

$$gl(1 - \cos(\theta_{\max})) = \frac{1}{2} \dot{\theta}^2 l^2 \quad \Bigg| \quad T = 2\pi \left(\frac{l}{g} \right)^{1/2}$$

$$g(1 - \cos(\theta_{\max})) = \frac{1}{2} \dot{\theta}^2 l$$

$$\frac{1}{2} \dot{\theta}^2 l^2 - gl(1 - \cos(\theta)) = -gl(1 - \cos(\theta_{\max}))$$

$$\frac{1}{2} \dot{\theta}^2 + \frac{g}{l} (\cos(\theta) - \cos(\theta_{\max})) = 0$$

$$- \sqrt{\frac{2g}{l}} \sqrt{\cos(\theta) - \cos(\theta_{\max})} = \frac{d\theta}{dt}$$

$$T = 4 \sqrt{\frac{l}{2g}} \int_0^{\theta_{\max}} \frac{d\theta}{\sqrt{\cos(\theta) - \cos(\theta_{\max})}}$$

$$\int \sqrt{\frac{\theta_{\max}^2}{2} - \frac{\theta^2}{2}} d\theta$$

$$= 4\sqrt{\frac{\ell}{g}} \int_0^{\theta_{\max}} \frac{d\theta}{\theta_{\max} (1 - \theta^2/\theta_{\max}^2)^{1/2}}$$

$$\sin^2 + \cos^2 = 1$$

$$\frac{\theta}{\theta_{\max}} = \sin(\varphi)$$

$$d\theta = \theta_{\max} \cos(\varphi) d\varphi$$

$$4\sqrt{\frac{\ell}{g}} \int_0^{\theta_{\max}} \frac{\cos(\varphi)}{\cos(\varphi)} d\varphi$$

$$= 4\sqrt{\frac{\ell}{g}} \sin^{-1}\left(\frac{\theta}{\theta_{\max}}\right) \Big|_0^{\theta_{\max}}$$

$$= 4\sqrt{\frac{\ell}{g}} \left(\sin^{-1}(1) - \sin^{-1}(0) \right)$$

$$\left(\frac{\pi}{2} - 0 \right)$$

$$2\pi\sqrt{\frac{\ell}{g}}$$

$$[P] = \frac{\text{kg m/s}^2}{\text{m}^2}, \quad \rho = \frac{\text{kg}}{\text{m}^3}$$

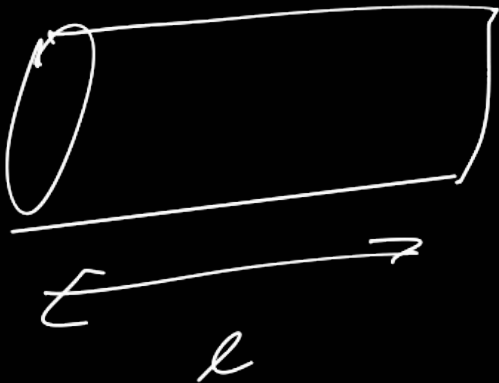
$$[P] = \frac{kg}{m s^2}$$

$$\left[\frac{P}{\rho} \right] = \frac{kg}{m s^2} \frac{m^3}{kg}$$

$$\frac{m^2}{s^2}$$

$$V^2 = K \frac{P}{\rho}$$

$$V_{air}^2 =$$



$$[X] \equiv \frac{Vol}{s}$$

$$P = \frac{F}{A}$$

$$[X] \sim \frac{[l A]}{\Delta t}$$

$$X(r, \frac{P}{\rho}, n) = \frac{m^3}{s}$$

$$\left[\frac{P}{\rho} \right] = \frac{m}{m^3} , [n] = N s m^{-2}$$

$$\frac{\left[\frac{P}{\rho} \right]}{[n]} \sim \frac{1}{N s m^3}$$

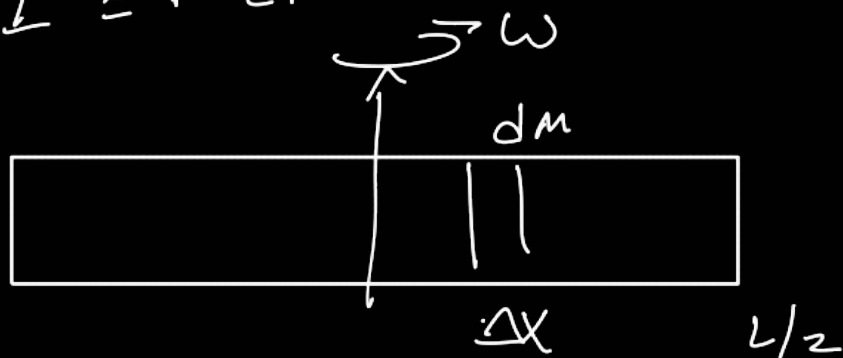
$$\frac{1}{m s}$$

$$K \left(\frac{r^2 \left(\frac{I}{r^2} \right)}{n} \right)$$

$$m = p \times$$

$$F = \frac{\Delta p}{\Delta t} = \frac{m \Delta v}{\Delta t}$$

$$dI = r^2 dm$$



$$\rho_L = \frac{M_{\text{tot}}}{L} = \frac{dm}{dx}$$

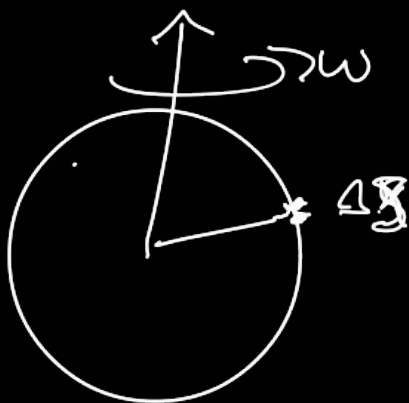
$$dm = \rho_L dx$$

$$\rho_L = \frac{M}{L}$$

$$I = 2 \int_0^{L/2} x^2 \rho_L dx$$

$$= \frac{2}{3} \rho_L \left(\frac{L}{2} \right)^3$$

$$= \frac{L^3}{12} \rho_L = \frac{ML^2}{12}$$



$$ds = R d\theta$$

$$\rho_L = \frac{dm}{ds}$$

$$dm = \rho_L ds$$

$$\int dI = \int r^2$$

$$dm = \frac{M R^2}{2\pi} \int_0^{2\pi} d\theta$$

$$= \underline{M R^2}$$

$$Mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$\omega = \frac{v}{R}$$

$$gh = \frac{1}{2}v + \frac{1}{2} R^2 \left(\frac{v^2}{R^2} \right)$$

$$\boxed{gh = v^2}$$

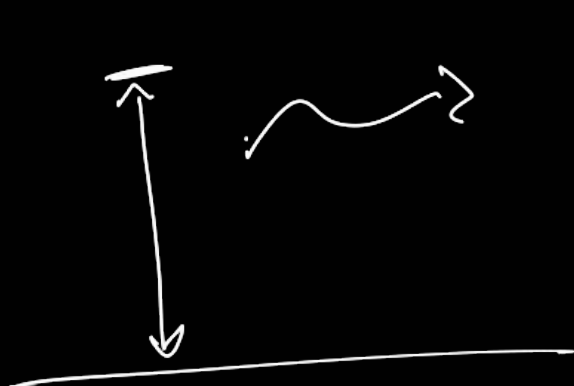
$$* V_B^2 = gh$$

$$V_B'^2 = g(0.9h) + V_C^2$$

$$V_B'^2 - g(0.9h) = V_C^2$$

$$\therefore 1gh = V_C^2$$

$$V_D = V_B$$



$$E_n = \frac{hc}{\lambda}$$

$$\Delta E = \frac{hc}{\Delta \lambda}$$

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

$$\Delta X = c \Delta t$$

$$\frac{\hbar c}{\lambda^2} \Delta \lambda \Delta t \geq \frac{\hbar}{2}$$

$$\Delta t \geq \frac{\hbar \lambda^2}{2 \hbar c \Delta \lambda}$$

$$\psi(x) = ()$$

$$\langle \psi | \psi \rangle = 1$$

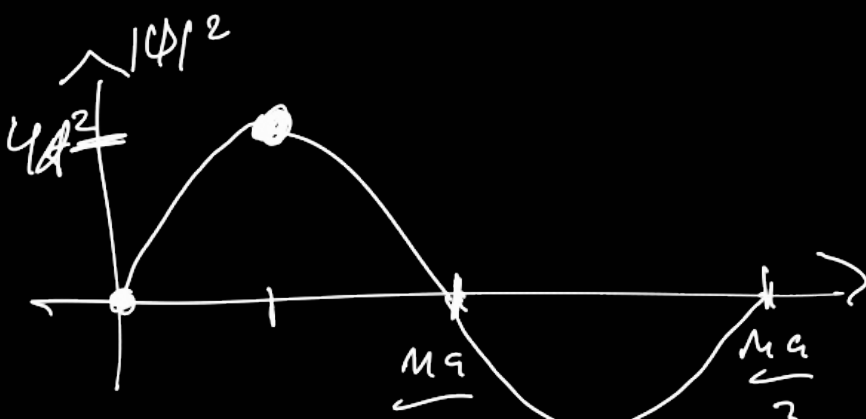
$$= \int |\psi|^2 dx$$

$$|\psi|^2 = \psi^* \psi = A^2 \left(e^{i \frac{8\pi x}{a}} - e^{-i \frac{4\pi x}{a}} \right) \left(e^{-i \frac{8\pi x}{a}} - e^{i \frac{4\pi x}{a}} \right)$$

$$= A^2 (1 - e^{-12i} - e^{12i} + 1)$$

$$= A^2 (2 - (e^{12i} + e^{-12i}))$$

$$= 2A^2 \left(1 - \cos\left(\frac{12\pi x}{a}\right) \right)$$



$$\frac{12\pi x}{a} = 2\pi m$$

$$x = \frac{ma}{6}$$

$$\sin\left(\frac{12\pi x}{a}\right) = 0$$

$$\frac{12\pi x}{a} = n\pi$$

$$\frac{na}{12}$$

$$\pi \cdot n$$

$$(1 - (-1)^n)$$

$$P_{\text{peak}} = \int_0^{a/4} 2A^2(1 - \cos(\gamma)) dx$$

